Semiconductor Analysis Toolkit - Help Guide

Overview

This GUI-based toolkit analyzes I-V characteristics and surface state densities in semiconductors using Thermionic and Cheung methods.

System Requirements

- Python >= 3.8
- Required Libraries: numpy, pandas, matplotlib, scipy, python-docx, openpyxl

How to Launch

Run the script:

python 123.py

Modules

1. I-V Analyzer
2. Surface State Analyzer

(Switch between using the navigation buttons at the bottom)

I-V Analyzer Instructions

1. Load your CSV/Excel file.
2. Select Voltage and Current column indices (1-based).
3. Set Temperature (K), Vmin, Vmax.
4. Choose smoothing method (optional).
5. Select analysis method: Thermionic or Cheung.
6. Click Analyze to view results and plots.
7. Use "Best Exponential Region" to auto-detect best fit range.
8. Click "Save to Word" to export report.

Surface State Analyzer Instructions

1. Load your CSV/Excel file.
2. Select column indices (Voltage and Current).
3. Input device parameters: I0, Phib, i, s, W_D, , T.

4. Click "Calculate Nss".

5. Click "Export Results" to save output.

Pre-Defined Constants

- q (Electron Charge): 1.602e-19 C
- k (Boltzmann Constant): 1.381e-23 J/K or 8.617e-5 eV/K
- A_eff (Effective Area): 1e-4 cm
- Richardson Constant A*:

n-type: 120 A/cmK

p-type: 32 A/cmK

Formulas Used

Thermionic Method:

- n = q / (slope x k x T)
- I0 = exp(intercept)
- Phib = (kT/q) x ln((A* x A_eff x T) / I0)
- ln(I) = slope x V + intercept
- Rs slope of dV/dlnI vs I

Cheung Method:

- dV/dlnI = Rs1 x I + n(kT/q)
- n = (slope1 x q) / (k x T)
- H(I) = V - (kT/q) x ln(I) = Rs2 x I + Phib
- Phib = intercept of H(I)
- I0 = exp(-q x Phib / (n x k x T))

Surface State Density:

- n(V) = V / (k_eV x T x ln(I/I0))
- Nss = (1/q) x [(i/) x (n - 1) - (s/W_D)]

(i, s in F/cm; , W_D in cm)

Troubleshooting

- File errors: Ensure valid numeric-only data.
- Rs mismatch: Indicates significant deviation in Rs fits.

- Plot issues: Check smoothing and column indices.
- Best Region requires at least 6 points.